Greenbrier Science Olympiad Invitational

Crime Busters B Test

2025-2026

Team Name:_____

Score:_____

The Crime:

On Wednesday afternoon at Greenbrier Middle School, several students were working in the STEM Makerspace to prepare for the upcoming STEM Expo. The students are competing to win a $50,000 scholarship. Sam was printing his Hand Grip Prototype in the 3D printer, which he made to help individuals with limited hand mobility grasp objects more easily. At the last meeting, Mr. James praised Sam for his design in front of his peers.

At 3:20, Sam's mom came to pick him up early to go to a dentist appointment. Rushing to leave, Sam left his prototype on the front workbench near the 3D printer. Between 3:25 and 4:05, Mr. James briefly stepped out to help another student who was working in the hallway. When he returned to put Sam's Hand Grip up, the prototype was gone.

The area showed some signs of a disturbance.
- A hex wrench on the floor beside the bench
- The rolling chair was repositioned with trace evidence on the wheel.
- Small fragments of plastic
- Visible powder residues on the workbench.
- a small liquid spill near the sink
- light fingerprint smudges on the storage bin and other places.

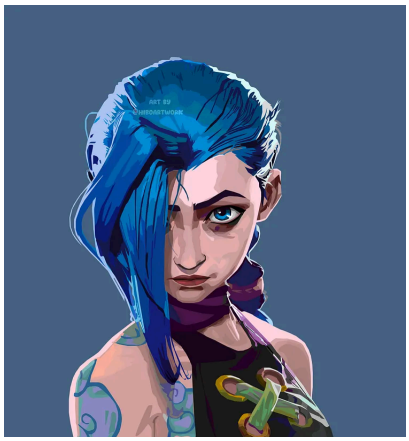
Five students were confirmed to be in the area or near the area during the period of time Mr. James was not in the room. Each had a legitimate reason to be present and had access to the workstations and tools near the workbench and 3D printer.



Vi

Vi is a hands-on student who often helps assemble or build large projects for the Robotics Club. She frequently wears athletic clothing made out of cotton, and she regularly handles storage bins and heavy tools that have plastic handles. Vi sometimes performs tests that involve powders such as salt and baking soda. Vi also walks through the school courtyard on her way to the Makerspace from her last period class, picking up bits of loam-type soil.

Statement to the Police: She was reorganizing tools and helping clean up the space before leaving. Vi has a plain whorl fingerprint.

 Jinx	Jinx is a creative but all-over-the-place designer who often experiments with powders, colorful plastics, and often uses the 3D printer. She rides her bike to school and parks her bike in the sandy path along the school building to get to the STEM wing. She often works quickly and leaves materials behind without realizing it. Jinx uses absorbent powders to clean sticky residues from her workstation. Statement to the Police: She was brainstorming a design idea and moving in and out of the room while testing parts. Jinx has an ulnar loop fingerprint.
 Jayce	Jayce is really competitive and focused on finishing his project for the expo. He primarily works with metals and various tools. Jayce parks his car in the student parking lot, which is paved. Jayce and Sam were previous project partners until they had a disagreement, which led them to split. He keeps a clean and organized workspace. Statement to the Police: He was checking progress on his part, he was printing on the other 3D machine, and also inspecting his presentation display. Jayce has a plain arch fingerprint.
 Caitlyn	Caitlyn is organized and pays attention to detail. She uses tools regularly. She constantly keeps her areas clean by using rubbing alcohol or hydrogen peroxide. She carries a fabric bag with her project notes and plastic containers with her powders in it. She often walks with Vi to the Markerspace through the courtyard path. Statement to the Police: She was helping Vi clean up. Caitlyn has a radial loop fingerprint.
 Ekko	Ekko is a resourceful builder who moves between the coding station and 3D printer areas. He often grabs miscellaneous tools and plastics for quick tests. He rides his scooter around campus and spends time near the STEM garden (soil is sandier) during lunch, where he plays with the school cat. He explained that he was testing the code of his upcoming project and may have moved some of the tools. Statement to Police: He does remember seeing Sam's prototype on the bench, but did not touch it. He claimed that he and Sam are best friends. Ekko has a tented arch fingerprint.

Qualitative Analysis

The forensics team has collected powders and mixtures at the scene. You will need to identify each of the powders, mixtures, and liquids in the table below

P1: was found on the workbench.

P2: was found in the trash can.

P3: was found on a paper towel near the sink.

P4: was found on the tool.

M1: was found in a small shared container used by multiple clubs. (2 powders mixed)

L1: Small spill collected from the sink area.

L2: Sample taken from a paper towel found in the trash.

L3: Residue collected from a handled tool.

1. P1:	**2.** P2:	**3.** P3:	**4.** P4:
5. M1:	**6.** L1:	**7.** L2:	**8.** L3:

9. What is the chemical formula for P1?:_____

10. Who does P1 implicate?:_____

11. What is the chemical formula for P2?:_____

12. Who does P2 implicate?:_____

13. What is the chemical formula for P3?:_____

14. Who does P3 implicate?:_____

15. What is the chemical formula for P4?:_____

16. Who does P4 implicate?:_____

17. What is the chemical formula for L1?:_____

18. Who does L1 implicate?:_____

19. What if the chemical formula for L2?:_____

20. Who does L2 implicate?:_____

21. What is the chemical formula for L3?:_____

22. Who does it implicate?:_____

23. What safety precautions should be taken when testing an unknown powder?_____

_____

24. Briefly describe the difference between the crystal size of sugar and salt.

_____

_____

_____

Polymer Testing

A. Plastics

Three plastic fragments were collected from the Makerspace. You have placed your results in the table below. Identify the plastic fragments by their full names and not their initials.

	Flame Behavior	Odor	Residue
P1: found near the prototype stand	Burns with a sooty flame	Sweet smell	Hard with a black bead
P2: found near the tool storage bin	Burns with a light blue flame and drips	Has a waxy and slightly paraffin smell	Soft with a cloudy bead
P3: collected near the sink	Burns with a steady yellow flame	Has a candle-like smell	white, crumbly ash/bead mixture

25. Identify Plastic 1:_____

26. Which suspect(s) does it implicate?:_____

27. Identify Plastic 2:_____

28. Which suspect(s) does it implicate?:_____

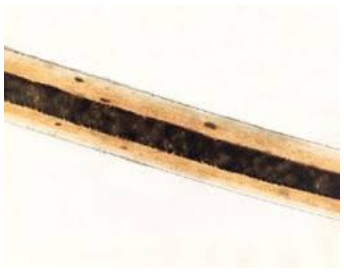
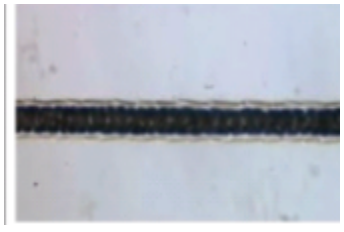
29. Identify Plastic 3:_____

30. Which suspect(s) does it implicate?:_____

What color flame does PVC have during a burn test?_____

B. Hair Evidence

Identify the following samples of hair that were found on the rolling chair.

 Hair Evidence 1	31. Identify Hair 1 and who does it implicate, if anyone? _____ _____
 Hair Evidence 2	32. Identify Hair 2 and who does it implicate, if anyone? _____ _____

33. How does the medulla differ between human hair and animal hair?

_____

_____

34. Why is hair considered class evidence instead of individual evidence?

_____

_____

C. Fiber Evidence

Identify the fiber that was collected from the scene based on the results of a burn test that was conducted and a picture that was taken of the fibers.

Fiber 1: Melts and burns, and produces a plastic-like smell and a hard bead. Fiber was collected from the handled tool 35. _____	
Fiber 2: Burns steadily and smells like burning paper, producing a soft gray ash. Fiber was collected from the workbench. 36._____	

Chromatography Testing

An ink sample was taken from a pen mark found on the workbench near the prototype. You will need to test the ink sample and calculate Rf values. Once completed, you will compare your results to the provided suspect chromatograms. Tape your evidence below and show RF calculations.

37. Which suspect's pen chromatogram best matches the crime scene sample?_____

38. What does Rf stand for, and why is it important when doing paper chromatography?

_____

39. Tape results below

Physical Evidence

A. Fingerprints

The following prints were lifted from the crime scene:

Lift 1: Was lifted from the hex wrench handle found by the prototype stand

Lift 2: Was lifted from the storage bin

Lift 3: Was lifted from the door

Lift 1	Lift 2	Lift 3
		

40. Lift 1 is what type of fingerprint?:_____

41. Lift 2 is what type of fingerprint?:_____

42. Lift 3 is what type of fingerprin?t:_____

43. Lift 1 implicates what suspect?_____

44. Lift 2 implicates what suspect?_____

45. Lift 3 implicates what suspect?_____

46. If a print is partial, what characteristics should you use to identify it?

_____

_____



Label the minutiae features for the following letters in the diagram

47. A. _____

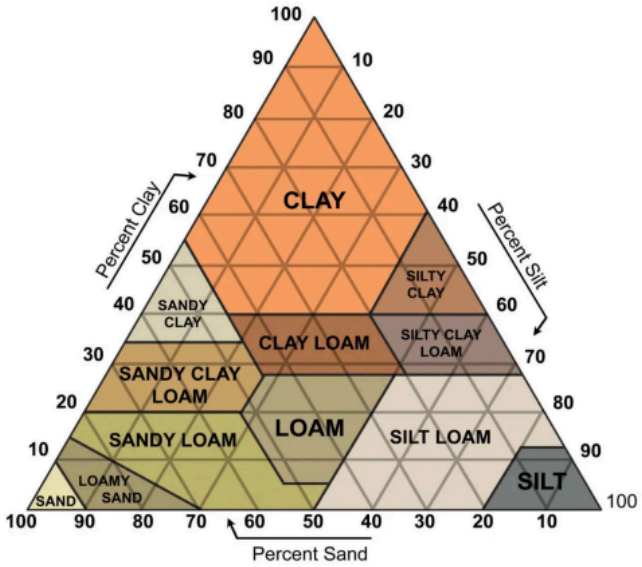
48 B. _____

49. C._____

50. D. _____

B. Soil Analysis

A soil sample was recovered from the area where the prototype went missing. The result of the soil sample is below.



? Sand	30% Silt	15% Clay

51. What would be the percentage of the sand?_____

52. What type of soil was found at the crime scene?_____

53. Which suspect(s) does this type of soil implicate?_____

54. Explain how particle size affects how soil behaves in water?

_____

_____

_____

Analysis

Using all the evidence collected from the Makerspace, determine which suspect most likely removed Sam's prototype.

State one suspect who is the culprit, and use at least four different evidence categories to help prove that they committed the crime. Explain how each piece supports your conclusion.

Also, use evidence to show how the other suspects are not responsible for committing the crime.

Greenbrier Invitational - Repaired practice edition

Division B | Crime Busters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (2 points) Authored written substitute for unavailable source material: An elongated stain's narrow tail points:

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

A. In a random direction

B. In the direction of travel

C. Back toward the source

D. Straight up

Answer: _____